



The Programming Life Cycle

Practical Skills Module 1: Topic 5

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Topic Code	PM-01-PS05:
Topic	Apply the programming life cycle to develop a solution

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0501 Apply the various stages (sequential and reiteratively) in the programming life cycle when developing an appropriate solution

Applied Knowledge

- AK0501 Concept, definition and functions of basic Python functionalities

Internal Assessment Criteria

- IAC0501 Expected results with the programming life cycle in Python are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-01



Practical task

Follow the facilitator's instructions to complete the following activities:

This section checks whether you understand how software is planned and improved using the programming life cycle. The activity is based on the provided **Weather Forecasting App** scenario, which requires you to explain how each stage is applied both step by step and iteratively in response to feedback, issues, or new needs.

Scenario: Developing a Weather Forecasting Mobile App

A team of developers is tasked with creating a mobile app that provides real-time weather forecasts. The app should display current weather conditions, hourly forecasts, and a 7-day outlook. Users should be able to search for locations, set preferences for units (e.g., Celsius or Fahrenheit), and receive alerts for severe weather.

The team must apply the **programming life cycle** to plan, design, build, test, and maintain the app. As part of this process, they will need to consider both sequential development (step-by-step) and iterative development (revisiting stages based on feedback, bugs, or new requirements).

- Create a document called: **LifeCycle.docx**

In the document, use the **Weather Forecasting App** scenario and explain the following seven stages:

- Problem Definition
- Feasibility Study and Planning
- System Design
- Implementation
- Testing
- Deployment
- Maintenance

For **each stage**, describe:

1. **Sequential application:** Explain how that stage is performed in a step-by-step development process.
2. **Iterative application:** Explain how that stage may be revisited and improved based on feedback, bugs, user needs, testing, or changing requirements.

After completing the document, convert it to PDF and save it as: **LifeCycle.pdf**

Also take a screenshot showing the completed document and save it as: **LifeCycle**. This requirement comes directly from the uploaded PM-01-PS05 task and must remain part of the capstone evidence.



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
